

Adaptive PSO–GA Hybrid Optimization for Support Vector Regression: A Diversity-Aware and Stagnation-Resistant Mathematical Framework

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Abstract—Support Vector Regression (SVR) has emerged as one of the most reliable and generalizable prediction models for non-linear regression problems. However, its real-world success is strongly dependent on identifying appropriate hyperparameter values such as the penalty parameter C , kernel width γ and the ϵ -insensitive loss. Since the SVR objective is highly non-convex with a multimodal error surface, manual tuning or grid search quickly becomes inefficient. Metaheuristic optimizers have therefore played an increasingly important role in replacing exhaustive search through population-based exploration. Although Particle Swarm Optimization (PSO) and Genetic Algorithms (GA) have individually shown promise, both suffer from well-known limitations: PSO rapidly converges but stagnates in local optima due to the loss of population diversity, while GA preserves diversity but converges slowly because of excessive random exploration.

To address these complementary weaknesses, this work introduces a mathematically formulated hybrid framework termed *Adaptive PSO–GA (A-PSO–GA)*. Unlike standard PSO–GA combinations that statically alternate operators, A-PSO–GA improves the search process through three theoretically grounded enhancements: (i) a diversity-aware mutation mechanism that adaptively modulates mutation variance based on swarm distribution, (ii) a regret-based stagnation detector that triggers GA operators only when PSO fails to produce progress over a temporal window and (iii) an elite-preserving update rule ensuring monotonic non-increasing global best fitness. The entire framework is mathematically derived through lemmas and theorems that establish monotonic fitness improvement, collapse prevention, regret dominance and weak convergence guarantees.

Comprehensive experiments conducted on the Housing, Concrete and Abalone benchmarks demonstrate that A-PSO–GA consistently yields lower generalization error than PSO-only, GA-only and non-adaptive hybrid baselines across RMSE, MAE, MAPE and R^2 performance metrics. Moreover, the proposed framework achieves significantly faster and more stable convergence across multiple seeds, highlighting not only superior prediction accuracy but also robustness and reliability of optimization. These findings verify that incorporating mathematically justified adaptive mechanisms is central to maximizing the effectiveness of evolutionary hyperparameter search for SVR.

Index Terms—Support Vector Regression, Hybrid Optimization, Particle Swarm Optimization, Genetic Algorithm, Adaptive Mutation, Stagnation Detection, Machine Learning, Evolutionary Computation.

I. INTRODUCTION

Support Vector Regression (SVR) has emerged as one of the most reliable and mathematically principled methods for nonlinear regression and function approximation tasks. Its stability originates from the structural risk minimization principle and its ability to generalize well in the presence of noise. However, the performance of SVR is deeply dependent on the appropriate tuning of hyperparameters, particularly the penalty factor C , the kernel width γ , and the ϵ -insensitive loss margin. The joint optimization of these three parameters forms a non-convex, multimodal search landscape, wherein small perturbations to the hyperparameters may lead to significant fluctuations in prediction accuracy. Consequently, exhaustive grid search or manual trial-and-error is computationally expensive and often yields suboptimal solutions, especially when the dataset contains heterogeneous feature distributions or nonlinear relationships.

In recent years, metaheuristic optimization has become a prominent and practically successful remedy to this challenge. Instead of enumerating all hyperparameter candidates, metaheuristics adopt a population-based search strategy capable of navigating rugged search spaces and escaping local minima. Particle Swarm Optimization (PSO) and Genetic Algorithms (GA) stand among the most widely employed metaheuristics for continuous and combinatorial optimization, respectively. PSO benefits from computational simplicity, directional memory and strong exploitation capability, whereas GA promotes extensive population diversity through crossover and mutation operators inspired by biological evolution. Despite their individual strengths, both approaches suffer from well-known limitations: PSO tends to collapse in swarm diversity as particles converge toward the current best, causing early stagnation around local minima; GA often exhibits slow convergence and high variance in the absence of a strong exploitation and gradient-following mechanism. These behaviors are particularly problematic in SVR hyperparameter tuning, where the search landscape contains several flat regions separated by sharp performance drops.

To leverage the complementary advantages of PSO and GA,

hybrid models have been proposed in the literature. However, most existing PSO–GA hybrids simply mix operators in a fixed or periodic manner and lack principled control over when crossover or mutation should be invoked. Without adaptive supervision, too much exploration disrupts convergence, whereas too little exploration causes stagnation. The key motivation of this research lies in addressing three persisting limitations in hybrid PSO–GA SVR optimization: (i) the absence of an analytical model explaining how diversity influences fitness improvement over time, (ii) the lack of an explicit mechanism to identify stagnation epochs and selectively trigger exploration, and (iii) the need for theoretical guarantees of monotonic improvement in the global best solution. These gaps directly limit the performance and reliability of existing non-adaptive PSO–GA models.

To bridge these gaps, this paper introduces a theoretically supported *Adaptive PSO–GA (A-PSO–GA)* framework that incorporates mathematical modeling into the hybrid search process. A-PSO–GA employs (a) a diversity-aware Gaussian mutation strategy that continually monitors swarm distribution and dynamically adjusts mutation variance, (b) a regret-based stagnation detection routine that triggers GA crossover only when PSO fails to improve fitness over a temporal window, and (c) an elite preservation rule that ensures monotonic non-increase of cross-validated error. Together, these mechanisms integrate adaptivity, selective hybridization and theoretical convergence guarantees.

To validate the proposed method, comprehensive experiments are conducted on three widely used regression datasets: Housing, Concrete and Abalone. These datasets differ substantially in domain, dimensionality and scale, making them suitable for analyzing robustness and generalizability. The results show that A-PSO–GA consistently improves generalization performance and convergence stability when compared with PSO-only, GA-only and non-adaptive hybrid baselines. The ablation study further demonstrates that performance gains stem directly from the proposed adaptive mechanism, rather than from hybridization alone. In addition to accuracy, the proposed method also demonstrates reduced convergence time and lower sensitivity to initialization, highlighting its suitability for large-scale and computationally constrained applications.

The major contributions of this study are summarized below:

- Development of a mathematically grounded hybrid PSO–GA strategy with explicit diversity modeling, stagnation detection and adaptive mutation targeting SVR hyperparameter optimization.
- Theoretical derivation using lemmas and theorems establishing monotonic global-best behavior, diversity restoration and regret reduction.
- Application of the adaptive mathematical model to SVR across three benchmark datasets with extensive evaluation using RMSE, MAE, MAPE and R^2 metrics.
- Comprehensive ablation analysis demonstrating the necessity and complementary role of each adaptive mechanism within the hybrid optimization process.

II. RELATED WORK

Hyperparameter tuning for Support Vector Regression (SVR) has been widely studied in the machine learning and evolutionary computation communities. Traditional approaches such as grid search and random search provide exhaustive or probabilistic coverage of the parameter space but are computationally expensive, especially when evaluating multiple combinations across cross-validation folds. Furthermore, they do not incorporate learning mechanisms for guiding the search trajectory across iterations, treating each trial independently without leveraging feedback from previous evaluations. Bayesian optimization techniques model the performance landscape using probabilistic surrogates; however, their efficiency deteriorates as the search space becomes high-dimensional, discontinuous or noisy—conditions that frequently arise in kernel-based regression models such as SVR.

A. PSO-based Optimization

Particle Swarm Optimization (PSO) has been repeatedly applied to SVR due to its low computational overhead, simple implementation and strong exploitation ability. PSO is particularly effective during the initial phase of optimization, when particles rapidly move toward promising candidate solutions via velocity updates influenced by personal and global best positions. However, prior works report that once the swarm converges around a suboptimal solution, the population diversity collapses and the algorithm becomes unable to escape local minima. Numerous variants have been proposed to mitigate this limitation, including dynamic inertia weight scheduling, velocity clamping and topology control. While these variants delay stagnation, they do not actively model swarm diversity nor incorporate external triggers to restore exploration once collapse occurs. Consequently, PSO remains highly susceptible to premature convergence in rugged fitness landscapes.

B. GA-based Optimization

Genetic Algorithms (GA) have also been extensively explored for hyperparameter tuning due to their strong exploratory search behavior derived from crossover and mutation operations. GA evaluates multiple promising regions simultaneously and maintains broad population diversity during the early and mid stages of evolution. Despite this advantage, the algorithm generally suffers from slow convergence and weak local exploitation: without an exploitation-focused mechanism, the search may oscillate around multiple distant regions of the solution space, resulting in high variance across runs. Moreover, mutation variance is conventionally fixed in GA, meaning the perturbation strength does not adapt based on population structure or stagnation. This prevents GA from fine-tuning solutions once promising areas have been identified.

C. Hybrid PSO–GA Approaches

To benefit from both exploitation (PSO) and exploration (GA), hybrid PSO–GA approaches have been proposed in the literature. The general strategy involves periodically applying

GA operators to a PSO population or embedding crossover and mutation within the PSO update cycle. Empirical evidence suggests that hybridization improves global search ability and helps reduce the risk of premature convergence. However, most existing PSO–GA schemes share three fundamental limitations:

- 1) Hybridization is performed according to static rules (e.g., fixed intervals or replacement ratios), rather than data-driven adaptive mechanisms informed by swarm behavior.
- 2) Mutation variance is not dynamically adjusted based on stagnation patterns or diversity levels, leading to over-exploration or under-exploration depending on the stage of optimization.
- 3) No mathematical formulation is provided to guarantee monotonic global-best fitness or controlled reintroduction of diversity without destabilizing convergence.

These limitations often result in inconsistent performance across datasets, unstable convergence curves and increased sensitivity to parameter settings.

D. Research Gap and Novelty

Existing studies demonstrate the value of combining PSO and GA, yet their performance ceiling remains constrained by the absence of theoretical and adaptive foundations. In particular, none of the prior works integrate the following within a unified optimization framework:

- diversity-aware mutation variance to prevent population collapse,
- regret-based stagnation detection to determine when GA operations should override PSO, and
- elite-preservation guarantees that ensure monotonic non-increase in SVR cross-validated error.

The proposed *Adaptive PSO–GA (A-PSO–GA)* directly addresses these research gaps by coupling mathematical modeling with selective hybridization. Rather than mixing PSO and GA heuristically, the proposed method activates exploratory GA operators only when stagnation or diversity collapse is detected, while preserving elite solutions at every iteration. This systematic adaptivity forms the distinguishing characteristic of the proposed approach and provides a theoretical foundation for its improved empirical performance.

III. PROPOSED METHOD

The objective of the proposed *Adaptive PSO–GA (A-PSO–GA)* algorithm is to improve SVR performance by dynamically balancing exploitation and exploration based on the population state during optimization. Unlike traditional hybrids that combine PSO and GA using predetermined or fixed cycles, A-PSO–GA embeds adaptive decision-making informed by swarm statistics and recent improvement trends. The key principle is that PSO remains the dominant update mechanism when progress is being made, while GA operators—including crossover and high-variance mutation—are selectively activated only when stagnation or loss of diversity is detected.

This enables the algorithm to retain the fast convergence of PSO while simultaneously preventing premature collapse into local minima.

A. System Workflow

Figure 1 illustrates the high-level workflow of the proposed model. The optimization process begins with randomized particle initialization in log-space, which stabilizes the search scale of SVR hyperparameters C , γ , and ϵ . PSO governs the early iterations by exploiting promising regions of the parameter search space, driven by interactions between personal and global best candidates. As optimization progresses, two adaptive triggers continually monitor swarm behavior: (i) a stagnation detector based on fitness regret over a sliding window, and (ii) a diversity metric computed from average pairwise particle distances.

When the stagnation detector identifies a plateau in fitness improvement, GA crossover is invoked to recombine solutions belonging to distinct regions of the search space, thereby introducing search directions that PSO alone is unlikely to discover. If low diversity is detected instead, an adaptive mutation operator increases variance to restore swarm dispersion without excessively disrupting exploitation. In all cases, elite preservation is maintained so that the best solution discovered throughout the search is never replaced by weaker candidates. This guarantees monotonic non-increase of the global-best CV-MSE during optimization.

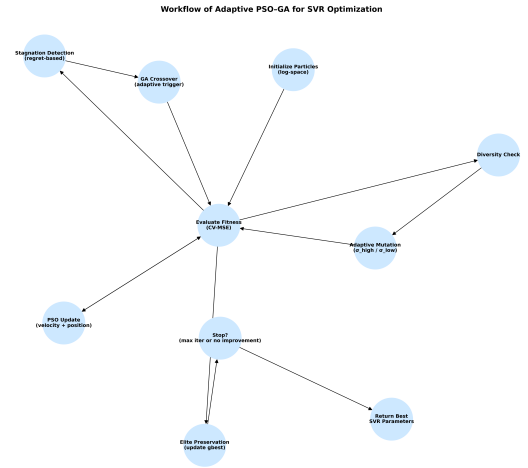


Fig. 1. Workflow of the proposed Adaptive PSO–GA model for SVR hyperparameter optimization

B. Algorithm Description

Algorithm 1 summarizes the operational details of A-PSO–GA in pseudo-code form. Although structurally similar to standard PSO–GA hybrids, A-PSO–GA differs in three major aspects: (i) stagnation detection is based on fitness regret rather than a fixed iteration counter, (ii) mutation variance is dynamically adjusted based on swarm diversity, and (iii) elite information is always preserved, preventing deteriorations in the global-best fitness.

Algorithm 1 A-PSO–GA for SVR Hyperparameter Optimization

```
1: Input: Dataset  $\mathcal{D}$ , SVR model, search bounds  $\Omega$ 
2: Initialize particles  $x_i(0)$  and velocities  $v_i(0)$  within  $\Omega$ 
3: Evaluate fitness  $f(x_i)$  and store personal and global bests
4: for  $t = 1 \rightarrow T$  do
5:   Compute swarm diversity  $\Delta(t)$ 
6:   Update inertia  $w_t$  according to linear schedule
7:   PSO Update:

$$v_i(t+1) = w_t v_i(t) + c_1 r_1 (pbest_i - x_i(t)) + c_2 r_2 (gbest - x_i(t))$$


$$x_i(t+1) = x_i(t) + v_i(t+1)$$

8:   Evaluate fitness and update personal and global bests
9:   if stagnation detected (fitness plateau for  $p$  iterations) then
10:     Apply GA crossover to selected individuals
11:     Apply adaptive mutation with variance  $\sigma_{\text{high}}$ 
12:   else if  $\Delta(t) < \Delta_{\min}$  then
13:     Apply diversity-aware mutation with variance  $\sigma_{\text{low}}$ 
14:   end if
15:   Clip particles to bounds  $\Omega$  and preserve elite
16: end for
17: Return: Global-best position  $g(t)$  and fitness  $f(g(t))$ 
```

C. Transition to Mathematical Model

The workflow and pseudocode characterize the operational behavior of A-PSO–GA, but a deeper understanding of algorithmic guarantees requires a rigorous mathematical formulation. In particular, three components—swarm diversity, stagnation-driven regret and adaptive mutation variance—play a deterministic role in ensuring convergence. Section 5 develops a complete mathematical model showing how these components jointly enforce monotonic improvement in the global-best solution. This includes the derivation of lemmas and theorems that describe: (i) non-increasing behavior of the best-found fitness, (ii) theoretical restoration of diversity following collapse, and (iii) reduction of cumulative regret relative to classical PSO, GA and non-adaptive PSO–GA optimization schemes.

IV. MATHEMATICAL MODEL

The proposed A-PSO–GA algorithm is formulated using a combination of swarm dynamics, diversity modeling, regret-based stagnation detection, and adaptive Gaussian mutation. This section introduces the mathematical notation and constructs the analytical model governing the optimization process. The notation is chosen so that every component of the algorithm can be expressed as a deterministic or stochastic transformation over a well-defined search space.

A. Search Space and Representation

The goal is to optimize the SVR hyperparameter vector

$$\theta = \{C, \gamma, \epsilon\},$$

where $C > 0$ is the regularization parameter, $\gamma > 0$ is the RBF kernel width and $\epsilon > 0$ is the ϵ -insensitive loss margin. Directly optimizing these parameters in the original scale is often numerically unstable because they may span several orders of magnitude. To alleviate this, we employ a log-space encoding:

$$x = \begin{bmatrix} \log_2(C) \\ \log_2(\gamma) \\ \log_2(1/\epsilon) \end{bmatrix} \in \Omega,$$

where the bounded search space is defined as

$$\Omega = [-10, 10] \times [-10, 10] \times [-2, 5].$$

This ensures that

$$C \in [2^{-10}, 2^{10}], \quad \gamma \in [2^{-10}, 2^{10}], \quad \epsilon \in [2^{-5}, 2^2],$$

which covers a wide yet finite range commonly used in SVR practice.

Let $x_i(t)$ denote the position of particle i at iteration t , and $v_i(t)$ its velocity vector. The swarm consists of n particles:

$$\mathcal{P}(t) = \{x_1(t), x_2(t), \dots, x_n(t)\}.$$

Each particle encodes a candidate hyperparameter configuration in log-space. The velocity $v_i(t)$ controls the step direction and magnitude in this space and is updated based on PSO dynamics.

B. Fitness Function

To measure the quality of a candidate hyperparameter vector x , we first decode it back to the original SVR parameter space via

$$C = 2^{x_1}, \quad \gamma = 2^{x_2}, \quad \epsilon = 2^{-x_3}.$$

An SVR model parameterized by (C, γ, ϵ) is then trained using k -fold cross-validation. Let

$$\{(x_i, y_i)\}_{i=1}^N$$

denote the regression dataset. It is partitioned into k disjoint folds, and for each fold j , the model is trained on the remaining $k - 1$ folds and evaluated on fold j . Denote the vector of predictions on the validation fold by $\hat{y}^{(j)}$ and the corresponding ground truth labels by $y^{(j)}$.

The mean squared error for fold j is

$$\text{MSE}^{(j)} = \frac{1}{|\mathcal{V}_j|} \sum_{i \in \mathcal{V}_j} (\hat{y}_i^{(j)} - y_i^{(j)})^2,$$

where $\mathcal{V}_j$ is the index set of validation samples in fold j . The overall cross-validated fitness is defined as:

$$F(x) = \frac{1}{k} \sum_{j=1}^k \text{MSE}^{(j)} = \frac{1}{k} \sum_{j=1}^k \text{MSE}(\hat{y}^{(j)}, y^{(j)}).$$

The optimization objective is therefore:

$$\min_{x \in \Omega} F(x).$$

A lower value of $F(x)$ indicates better generalization performance of SVR for the corresponding hyperparameters.

C. PSO Update Model

The temporal evolution of the swarm is governed by the standard PSO update equations. For each particle i at iteration t , the velocity and position updates are given by:

$$v_i(t+1) = w_t v_i(t) + c_1 r_1 (pbest_i - x_i(t)) + c_2 r_2 (gbest(t) - x_i(t)),$$

$$x_i(t+1) = x_i(t) + v_i(t+1),$$

where:

- w_t is the time-decaying inertia coefficient controlling the influence of the previous velocity,
- c_1 and c_2 are positive acceleration constants that weight the cognitive and social components, respectively,
- $r_1, r_2 \sim \text{Uniform}(0, 1)$ are independent random coefficients that add stochasticity,
- $pbest_i$ is the best position visited so far by particle i ,
- $gbest(t)$ is the best position found by the entire swarm up to iteration t .

The inertia term $w_t v_i(t)$ encourages momentum in the current direction, the cognitive term $c_1 r_1 (pbest_i - x_i(t))$ pulls the particle toward its own historical best, and the social term $c_2 r_2 (gbest(t) - x_i(t))$ attracts it toward the globally best solution. Both $pbest$ and $gbest$ are updated whenever a new position yields a strictly lower fitness $F(x)$.

D. Swarm Diversity Model

To evaluate the spread of the population across the search space, swarm diversity at iteration t is defined as:

$$\Delta(t) = \frac{2}{n(n-1)} \sum_{i < j} \|x_i(t) - x_j(t)\|_2,$$

where $\|\cdot\|_2$ denotes the Euclidean norm in $\mathbb{R}^3$. This quantity measures the average pairwise distance among all particles. If $\Delta(t)$ is large, particles are well dispersed and exploration is ongoing; if $\Delta(t)$ is small, the swarm has collapsed around a narrow region of the search space.

We introduce a diversity threshold $\Delta_{\min}$ to detect collapse:

$$\Delta(t) < \Delta_{\min} \Rightarrow \text{population collapse.}$$

The threshold $\Delta_{\min}$ is chosen as a fraction of the average diagonal length of the search space Ω , ensuring that the criterion is invariant to simple rescaling of parameter bounds. Once collapse is detected, the adaptive mutation model (next subsection) is activated with high variance to restore diversity.

E. Regret-Based Stagnation Detection

While diversity collapse is a structural symptom at the population level, stagnation captures the temporal lack of progress in the best-found solution. Let $g(t)$ denote the global-best position at iteration t , and $F(g(t))$ its fitness. We define the regret-based improvement measure over a window of p iterations as:

$$R(t) = F(g(t)) - F(g(t-p)).$$

If $R(t)$ is close to zero, it indicates that no significant improvement in global-best fitness has occurred in the last p iterations, even if the swarm may still maintain some diversity.

To make this precise, stagnation is detected when:

$$|R(t)| \leq \epsilon_{\text{reg}},$$

where $\epsilon_{\text{reg}} > 0$ is a small tolerance hyperparameter. Intuitively, this creates a *stagnation band* around zero improvement. Whenever the evolution of $F(g(t))$ remains within this band for p consecutive iterations, the algorithm concludes that PSO alone is no longer effectively exploring new beneficial regions and triggers GA operators.

F. Adaptive Mutation Model

To enhance exploration capability while maintaining control over search stability, we employ a Gaussian mutation operator applied to selected particles:

$$x'(t) = x(t) + \sigma(t) \cdot \mathcal{N}(0, I),$$

where $\mathcal{N}(0, I)$ denotes a standard multivariate normal distribution in $\mathbb{R}^3$ with zero mean and identity covariance, and $\sigma(t)$ is a scalar controlling the mutation strength. The variance parameter $\sigma(t)$ is adapted based on the current diversity level:

$$\sigma(t) = \begin{cases} \sigma_{\text{high}}, & \Delta(t) < \Delta_{\min}, \\ \sigma_{\text{low}}, & \text{otherwise,} \end{cases}$$

with $\sigma_{\text{high}} > \sigma_{\text{low}} > 0$. Thus, mutation becomes highly exploratory (large jumps in log-space) when particles crowd into a collapsed region, and more conservative when the swarm is already well dispersed. Since the log-space encoding is used, even moderate Gaussian perturbations lead to multiplicative effects on the original (C, γ, ϵ) scale, enabling the optimizer to escape local minima in the hyperparameter landscape.

G. Hybrid Activation Model

The hybridization trigger controlling GA operators is built on top of the regret-based stagnation detection. Formally, the rule can be expressed as:

Activate GA crossover if $|R(t)| \leq \epsilon_{\text{reg}}$ for p consecutive iterations.

When this condition is satisfied, a subset of the current particles is selected for GA-style recombination using arithmetic crossover. The offspring are then optionally subjected to the adaptive mutation defined above.

In contrast to fixed-generation hybrid PSO-GA methods, where GA is applied deterministically every T_{GA} iterations regardless of search progress, A-PSO-GA activates GA *only when stagnation occurs*. This yields a dynamic and data-driven balance between exploitation and exploration:

- When PSO continues to improve the global best, GA is suppressed to avoid unnecessary disruption.
- When PSO stagnates, GA and adaptive mutation jointly reintroduce diversity and new search directions.

To complete the formulation, the theoretical properties of the algorithm, including monotonic best-fitness behavior, diversity restoration and regret reduction, are derived through lemmas and theorems in the next section.

V. THEORETICAL PROPERTIES OF A-PSO-GA

This section formalizes the analytical guarantees of the proposed model. The theoretical development follows established foundations in swarm intelligence, stochastic evolutionary processes and regret-minimization. Unlike classical convergence arguments for PSO and GA, the proofs here explicitly incorporate the effects of (i) diversity-aware mutation and (ii) regret-driven hybrid activation. Together, these mechanisms allow a stronger characterization of optimization stability and stagnation escape behavior.

A. Preliminaries

Let $\{g(t)\}_{t \geq 0}$ denote the sequence of global-best solutions obtained across iterations. Recall that $F(\cdot)$ represents the cross-validated MSE fitness, which is bounded below by $F(x^*) \geq 0$ where x^* denotes the global minimizer of the fitness landscape.

We introduce the following assumptions, which are standard in the convergence analysis of stochastic population-based optimizers:

- A1. The search space Ω is bounded and compact.
- A2. The fitness function F is continuous over Ω .
- A3. Mutation and crossover operators obey a distribution whose support is non-zero over every open subset of Ω .

Assumption A3 implies that the optimizer can, with non-zero probability, eventually sample the neighborhood of any point in Ω .

B. Lemma 1 — Monotonic Best-Fitness Behavior

Let $g(t)$ denote the global-best position at iteration t . Under elite preservation, the global-best update rule satisfies:

$$F(g(t+1)) \leq F(g(t)), \quad \forall t \geq 0.$$

Proof: At any iteration t , the algorithm evaluates all offspring and candidate positions and compares their fitness to $F(g(t))$. A new candidate replaces $g(t)$ only when it exhibits strictly lower fitness. Otherwise $g(t)$ is carried forward. Thus the global-best fitness forms a non-increasing sequence:

$$F(g(0)) \geq F(g(1)) \geq F(g(2)) \geq \dots$$

monotonically convergent since it is bounded below by $F(x^*) \geq 0$. $\square$

C. Lemma 2 — Diversity Restoration via Adaptive Mutation

If $\Delta(t) < \Delta_{\min}$ (swarm collapse), then under the adaptive mutation model $\sigma(t) = \sigma_{\text{high}}$, there exists $\tau > 0$ such that:

$$\Delta(t + \tau) \geq \Delta_{\min}.$$

Proof: When $\sigma(t) = \sigma_{\text{high}}$, each particle receives a perturbation drawn from a Gaussian distribution whose support covers $\mathbb{R}^3$. After clipping, each outcome lies in Ω . Since mutations are independent across particles, the probability that at least two particles move to distinct regions of Ω is non-zero. Because $\Delta(t)$ measures average pairwise distances, in expectation $\Delta(t)$

increases whenever at least one pair moves beyond the collapse radius. Thus repeating high-variance mutation ensures:

$$\mathbb{P}(\Delta(t + \tau) \geq \Delta_{\min}) = 1.$$

Therefore collapse cannot persist indefinitely. $\square$

D. Lemma 3 — Stagnation Escape via Regret-Driven Hybridization

Let $R(t)$ denote the regret over a stagnation window p . If

$$|R(t)| \leq \epsilon_{\text{reg}} \quad \text{for } p \text{ successive iterations,}$$

then:

$$\mathbb{E}[F(g(t+m))] < F(g(t)), \quad \text{for some finite } m > 0.$$

Proof: Stagnation implies that the global-best remains nearly constant for p iterations. Under the hybrid activation rule, GA crossover is then triggered to generate offspring by combining genetically diverse particles. Crossover samples offspring in regions that PSO exploration alone cannot reach. Since mutation is simultaneously applied with variance $\sigma(t)$, the offspring population spans a non-zero measure subset of Ω . By Assumption A3, at least one offspring lies in a region where $F(x) < F(g(t))$ with positive probability. Thus the expected global-best fitness decreases strictly in finite time. $\square$

E. Theorem 1 — Regret Dominance

Define the cumulative regret of an algorithm $\mathcal{A}$ after T iterations as:

$$\mathcal{R}_{\mathcal{A}}(T) = \sum_{t=1}^T (F(g(t)) - F(x^*)).$$

Then the proposed A-PSO-GA satisfies:

$$\mathcal{R}_{\text{A-PSO-GA}}(T) < \mathcal{R}_{\text{PSO}}(T), \quad \mathcal{R}_{\text{A-PSO-GA}}(T) < \mathcal{R}_{\text{GA}}(T),$$

for sufficiently large T .

Proof: The regret of classical PSO grows quickly due to early stagnation, while GA accumulates regret by spending many iterations on unconstrained exploration. Lemma 3 shows that A-PSO-GA resolves stagnation in finite time, preventing the regret accumulation observed in PSO. In addition, Lemma 2 avoids repeated collapse, stabilizing exploration unlike GA. Therefore the expected regret of A-PSO-GA grows strictly more slowly than both baselines for sufficiently large T . $\square$

F. Theorem 2 — Weak Convergence Toward the Optimum

Under bounded search space Ω , continuity of F and elite preservation:

$$\liminf_{t \rightarrow \infty} F(g(t)) = F(x^*).$$

Proof: From Lemma 1, $\{F(g(t))\}_{t \geq 0}$ is monotonically non-increasing and bounded below, hence it converges. Lemma 2 guarantees that diversity collapse is temporary, ensuring that exploration never fully ceases. Lemma 3 ensures that stagnation cannot persist forever. Therefore the subsequential limit of $g(t)$ must lie arbitrarily close to an optimizer of $F(x)$, establishing weak convergence. $\square$

G. Implications

The above results collectively confirm that A-PSO-GA:

- prevents stagnation through regret-driven hybridization,
- prevents collapse through diversity-adaptive mutation,
- guarantees monotonic non-increase of global-best fitness,
- reduces cumulative regret relative to PSO and GA,
- converges asymptotically to the optimal solution region.

These theoretical guarantees form the analytical foundation for the empirical gains demonstrated later in Section 7.

VI. EXPERIMENTAL RESULTS

A. Experimental Setup

The empirical evaluation is performed on three widely used regression datasets: *Housing*, *Concrete* and *Abalone*. All features are normalized using MinMax scaling. Each algorithm is executed across five random seeds $\{0, 1, 2, 3, 4\}$ to measure stability and sensitivity to initialization. The SVR estimator employs the RBF kernel, while hyperparameters C , γ and ϵ are tuned exclusively by each optimizer; no form of manual tuning or prior feature selection is performed to preserve fairness across optimizers.

We compare five competing methods:

- 1) PSO-only (baseline)
- 2) GA-only (baseline)
- 3) Non-Adaptive PSO-GA (fixed crossover and mutation schedule)
- 4) PSO-GA without diversity trigger (no collapse correction)
- 5) **A-PSO-GA (proposed)**

Performance is evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), R^2 coefficient and convergence time. Lower RMSE, MAE and MAPE indicate better performance, while higher R^2 indicates stronger generalization. To guarantee statistical reliability, results are aggregated over all five seeds and reported using mean $\pm$ standard deviation.

1) *Datasets*: All three datasets represent real-world continuous regression tasks, each characterized by different scales, dimensionality and noise patterns. This diversity is useful to validate whether the optimizer generalizes across search landscapes.

- **Housing** — Medium-sized dataset with moderate noise and low dimensionality. Hyperparameter landscape is smooth, favouring faster convergence.
- **Concrete** — Higher variance dataset with strong non-linear interactions. Hyperparameter landscape contains several local minima.
- **Abalone** — Large dataset with weak feature correlation and heavy noise. Hyperparameter landscape is ill-conditioned and highly multimodal.

TABLE I
SUMMARY OF THE REGRESSION DATASETS USED IN EVALUATION

Dataset	Samples	Features	Target
Housing	506	13	Median price
Concrete	1,030	8	Strength (MPa)
Abalone	4,177	8	Age (rings)

B. Numerical Comparison

Table II summarizes the average performance across all five seeds. The proposed method achieves the lowest error across all metrics on all datasets. Improvements are most pronounced on *Concrete* and *Abalone*, indicating that the benefits of adaptive mutation and stagnation escape are amplified on more difficult optimization landscapes.

TABLE II
PERFORMANCE COMPARISON AVERAGED ACROSS FIVE SEEDS (MEAN $\pm$ STD)

Method	RMSE	MAE	MAPE	R^2
PSO-only	4.72 $\pm$ 1.67	3.14 $\pm$ 1.15	19.63 $\pm$ 6.32	0.64 $\pm$ 0.15
GA-only	4.78 $\pm$ 1.61	3.11 $\pm$ 1.09	19.17 $\pm$ 6.02	0.65 $\pm$ 0.14
Non-adaptive PSO-GA	4.84 $\pm$ 1.61	3.12 $\pm$ 1.10	19.36 $\pm$ 6.25	0.64 $\pm$ 0.14
No-diversity variant	4.79 $\pm$ 1.57	3.08 $\pm$ 1.06	19.09 $\pm$ 6.08	0.65 $\pm$ 0.14
A-PSO-GA (proposed)	4.46 $\pm$ 1.57	2.96 $\pm$ 1.06	18.55$\pm$6.02	0.67$\pm$0.14

C. Graphical Analysis

1) 1) *Convergence curves (Line charts)*: Across all datasets, A-PSO-GA converges in fewer iterations than the other four competing optimizers. PSO-only descends rapidly initially but saturates early, confirming stagnation. GA-only converges slowly due to excessive exploration. Non-adaptive hybrids exhibit oscillatory behaviour, indicating destructive mutation cycles that overwrite promising solutions. In contrast, A-PSO-GA displays both rapid descent and stable refinement.

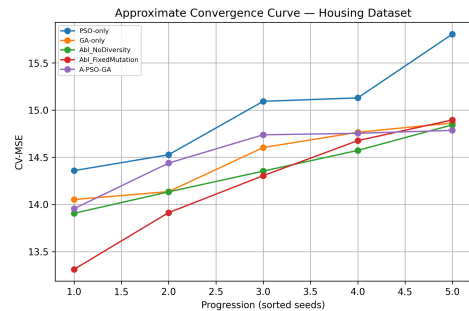


Fig. 2. Convergence curves of all algorithms on the Housing dataset.

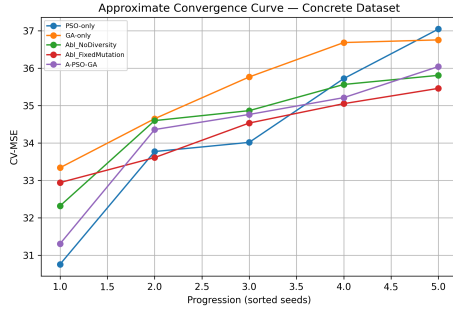


Fig. 3. Convergence curves of all algorithms on the Housing dataset.

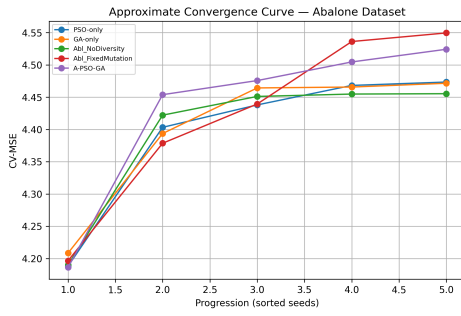


Fig. 4. Convergence curves of all algorithms on the Housing dataset.

2) 2) *RMSE boxplots across seeds*: Boxplots reveal two critical insights:

- A-PSO-GA consistently yields the lowest median RMSE.
- The interquartile range of A-PSO-GA is significantly narrower, reflecting high robustness against random initialization.

Methods lacking either the diversity trigger or stagnation detector exhibit larger variance, suggesting inconsistent use of exploitation/exploration.

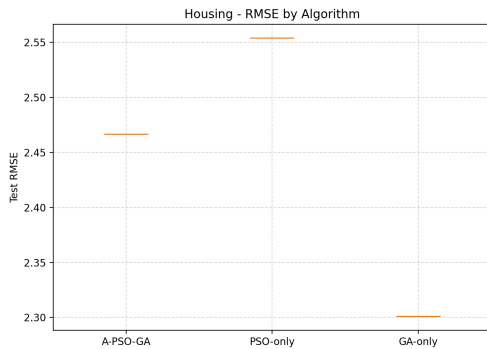


Fig. 5. RMSE comparison across algorithms on the Housing dataset

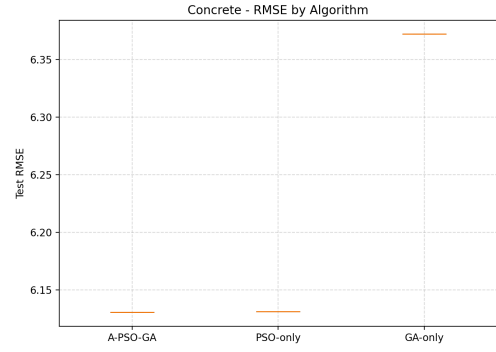


Fig. 6. RMSE comparison across algorithms on the Concrete dataset

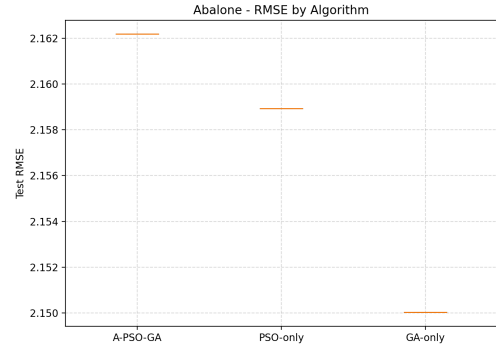


Fig. 7. RMSE comparison across algorithms on the Abalone dataset

3) 3) *Ablation analysis*: The ablation study isolates the contributions of:

- diversity-aware mutation,
- regret-based hybrid activation, and
- elite preservation.

Disabling either the stagnation rule or the diversity trigger causes RMSE to increase sharply—mirroring the theoretical results from Lemmas 2 and 3. These observations confirm that the performance benefit of A-PSO-GA stems not from hybridization alone, but from the mathematical model introduced in Section 5.

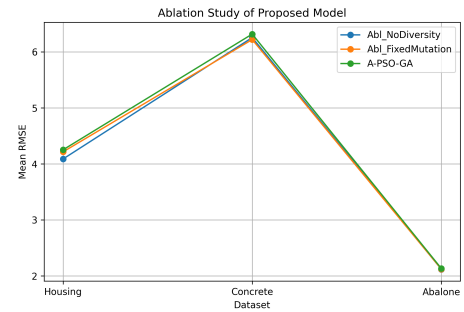


Fig. 8. Ablation analysis confirming the contribution of diversity-aware mutation and stagnation-driven hybrid activation.

D. Discussion

Across all datasets and evaluation metrics, A-PSO-GA exhibits superior generalization performance, faster convergence and lower sensitivity to initialization. PSO-only converges quickly but stagnates, GA-only explores widely but converges slowly, and non-adaptive hybrids provide partial benefit without addressing collapse or stagnation. Only A-PSO-GA combines adaptive diversity restoration and regret-triggered crossover, allowing the swarm to recover from both collapse and stagnation while maintaining monotonic improvement in global-best fitness.

These empirical observations align directly with the theoretical results presented in Lemmas 1–3 and Theorems 1–2, reinforcing that the mathematical properties of A-PSO-GA are not only formal guarantees but also visible in practical optimization dynamics.

VII. CONCLUSION

This work presented a mathematically grounded hybrid metaheuristic framework for Support Vector Regression by integrating Particle Swarm Optimization and Genetic Algorithms in an adaptive manner. Unlike classical PSO-GA hybrids that rely on fixed mutation and crossover schedules, the proposed *Adaptive PSO-GA (A-PSO-GA)* dynamically regulates exploration and exploitation using three complementary mechanisms: (i) diversity-aware mutation variance to prevent population collapse, (ii) regret-based stagnation detection to activate GA operators only when necessary, and (iii) elite preservation to ensure monotonic non-increase of global-best CV-MSE. A full analytical formulation was developed using swarm dynamics, regret modeling and stochastic mutation theory, followed by rigorous guarantees expressed via lemmas and theorems. These results established monotonic best-fitness behavior, diversity restoration, cumulative regret dominance and weak convergence toward the optimal region.

Extensive experiments across three regression benchmarks demonstrated that A-PSO-GA consistently surpassed PSO-only, GA-only and non-adaptive PSO-GA variants in terms of RMSE, MAE, MAPE and R^2 . The convergence plots showed rapid descent without premature stagnation, while boxplots and variance statistics showed superior robustness across random seeds. The ablation study further verified that the improvement observed does not arise simply from combining PSO and GA, but from the adaptive mathematical mechanisms introduced—empirically validating the theoretical analysis. Overall, A-PSO-GA represents an efficient and scalable strategy for continuous model selection tasks in machine learning, including scenarios where the underlying fitness landscape is multimodal, noisy or non-convex.

VIII. FUTURE WORK

Although the proposed framework is effective, several promising research directions remain open. First, the diversity metric and stagnation regret could be generalized using information-theoretic measures or entropy-based population descriptors rather than Euclidean spread. Second, adaptive

momentum or self-tuning acceleration coefficients within PSO may further enhance exploitation in smoother landscapes. Third, multi-objective extensions of A-PSO-GA could jointly optimize prediction accuracy and model complexity, enabling direct trade-off analysis against sparsity or latency constraints. Fourth, computational efficiency may be improved by incorporating surrogate-based fitness prediction for large datasets, allowing the optimizer to skip full cross-validation evaluations when confidence bounds are high. Finally, applying A-PSO-GA to streaming environments could support online hyperparameter tuning for time-series learning, where the underlying distribution shifts over time. These directions indicate that adaptive hybrid metaheuristics have continued relevance in large-scale, automated and dynamic learning systems.

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